# The wedge for Lecture 3

**If the product X is a necessity, like rice and flour, and the product Y is not a necessity, like perfume and sport equipment, then**

- **How does the quantity demanded change as the prices of the product X and Y change? Are the quantities of changes identical once the changes in prices are the same?**

- **How does the quantity supplied change as the prices of the product X and Y change? Are the quantities of changes identical once the changes in prices are the same?**

# Another scenario:

- You design websites for local businesses.
  - You charge $200 per website, and currently sell 12 websites per month.
- Your costs are rising (including the opportunity cost of your time)
  - You consider raising the price to $250.
- The law of demand: you won't sell as many websites if you raise your price.
  - How many fewer websites?
  - How much will your revenue fall, or might it increase?

# Lecture 3: 供需彈性與均衡分析

**(Elasticity and the Analysis of Equilibrium)**

# Outline

- 3-1
  - 3-1.1: The definition of elasticity
  - 3-1.2: Demand and Supply Curves and their Price Elasticities of Demand and Supply
  - 3-1.3: Two Special Cases for Demand and Supply Curves and their Price Elasticities
- 3-2
  - 3-2.1: The relation between elasticity and equilibrium
  - 3-2.2: The determinants of price elasticities
  - 3-2.3: Relation between price elasticity of demand and suppliers' total revenue
  - 3-2.4: Two examples

# 3-1.1: 供需彈性的定義

**(The Definition of Elasticity )**

# 變動量的比例

- 我們將<u>供需數量變動量比例</u>表示成:

$$e = \frac{\Delta Q}{\Delta Z}$$

(Responsiveness of quantity demanded $Q^D$ or quantity supplied $Q^S$ to a change in one of its determinants)

其中

(1) $Q$ 代表需求量或供給量

(Q is quantity demanded or quantity supplied)

(2) $Z$ 代表造成 $Q$ 變動的其他變數，例如價格

(Z the determinant of Q)

(3) $\Delta Q$ 和 $\Delta Z$ 代表 $Q$ 和 $Z$ 的變動量

($\Delta Q$ and $\Delta Z$ are the change volume of Q and Z)

# Responsiveness as elasticity

- **If $Q \equiv Q_X^D$ (the quantity demanded for product X), then it could be described as**

$$Q_X^D = f\,(P_X \mid P_Y, I, \dots).$$

**We define**

$$e_X^{D.P_X} = \frac{\Delta Q_X^D}{\Delta P_X} \text{ as the price elasticity of demand}$$

- **If $Q \equiv Q_X^S$ (the quantity supplied for product X), then it can be described as**

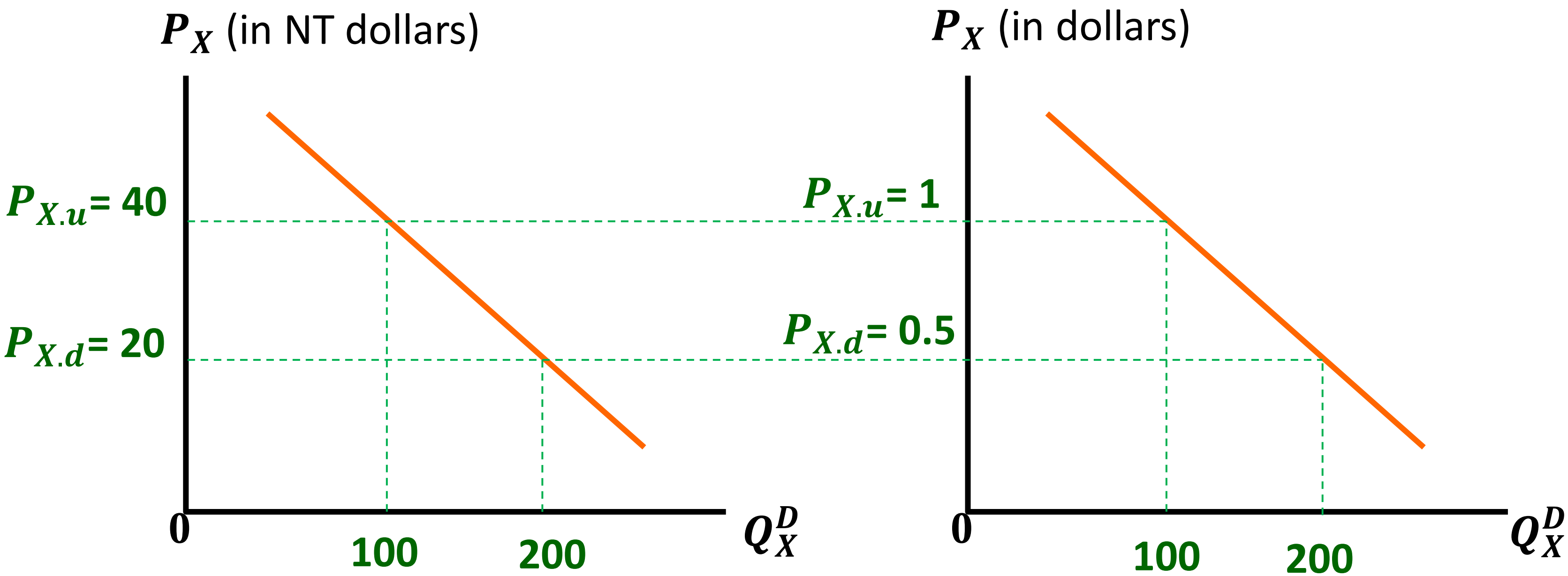
$$Q_X^S = g\,(P_X \mid P_Y, \dots),$$

**We define**

$$e_X^{S.P_X} = \frac{\Delta Q_X^S}{\Delta P_X} \text{ as the price elasticity of supply.}$$

# Lack of consistency

Dollar : NT Dollar = 1 : 40




- Same product and the same demand curve. All other things being equal but the price is measured in different currencies:



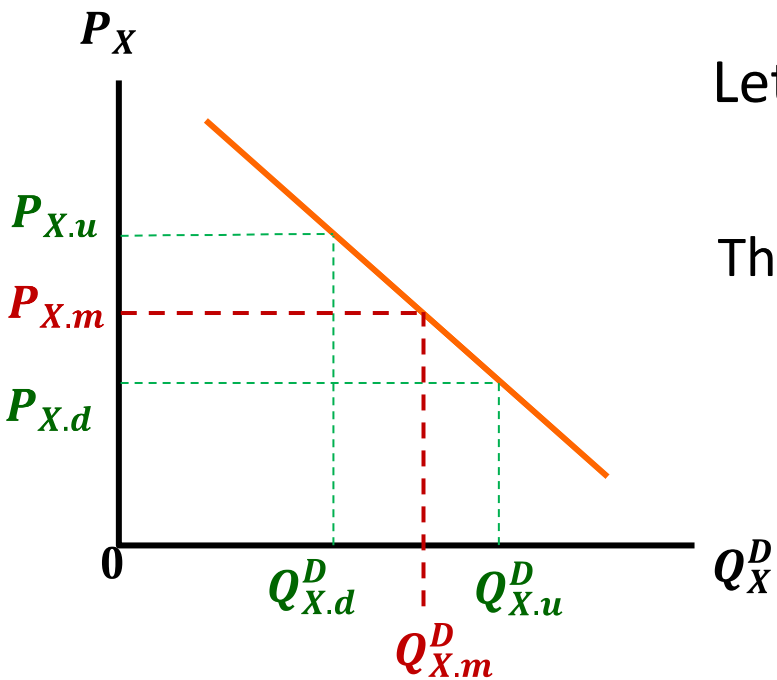
If the price is measured in NTD, then

$$e_X^{D.P_X} = \frac{\Delta Q_X^D}{\Delta P_{X.NTD}} = \frac{200 - 100}{20 - 40} = -5$$

If the price is measured in dollars

$$e_X^{D.P_X} = \frac{\Delta Q_X^D}{\Delta P_{X.D}} = -200$$

# Redefine the price elasticity of demand via midpoint method




Let $Q^D_{X.m} = \frac{Q^D_{X.u} + Q^D_{X.d}}{2}$, $P_{X.m} = \frac{P_{X.u} + P_{X.d}}{2}$

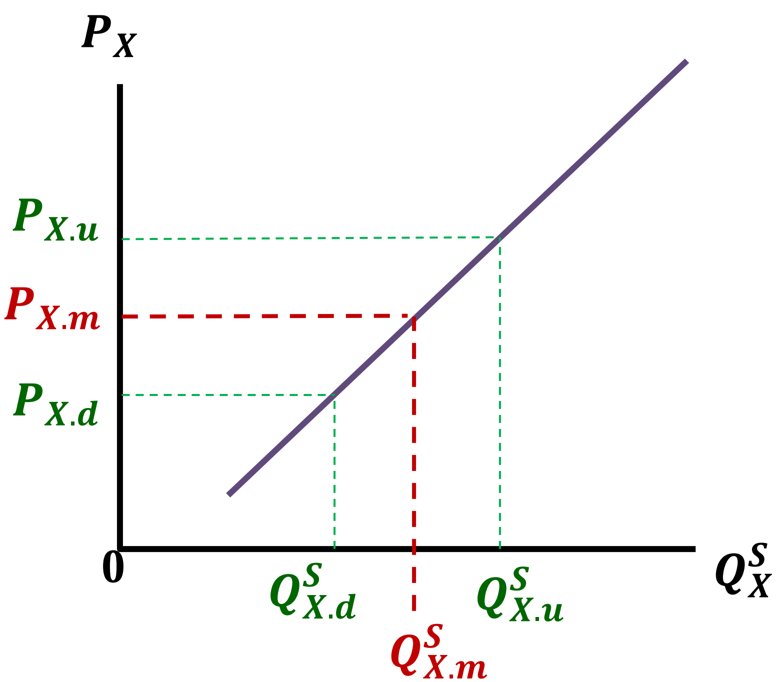
The price elasticity of demand is refined as

Percentage change of quantity demanded

$$E_X^{D.P_X} = \left| \frac{\Delta Q_X^D / Q^D_{X.m}}{\Delta P_X / P_{X.m}} \right| = \left| \frac{\Delta Q_X^D}{\Delta P_X} \right| \cdot \frac{P_{X.m}}{Q^D_{X.m}}$$

Percentage change of price

# Redefine the price elasticity of supply via midpoint method




Let $Q^S_{X.m} = \frac{Q^S_{X.u}+Q^S_{X.d}}{2}$, $P_{X.m} = \frac{P_{X.u}+P_{X.d}}{2}$

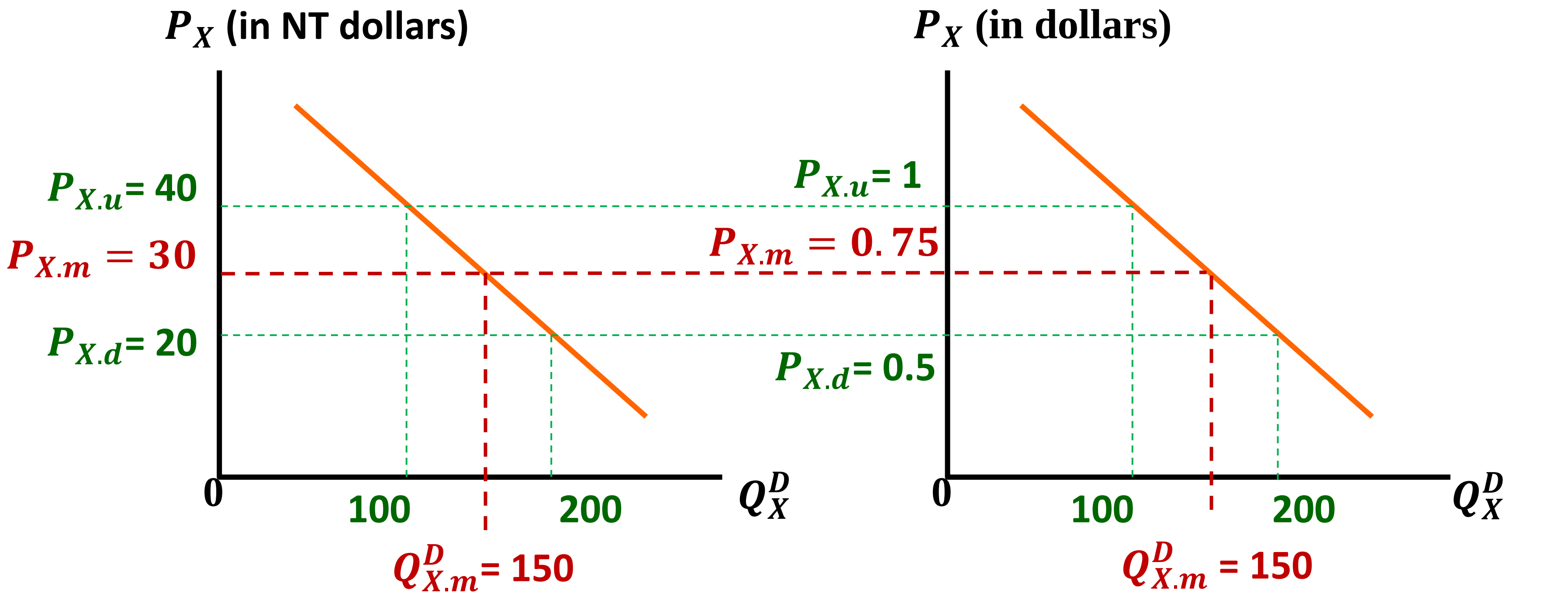
The price elasticity of supply is refined as

Percentage change of quantity supplied

$$E_X^{S.P_X} = \left| \frac{\Delta Q^S_X \big/ Q^{S.m}_X}{\Delta P_X \big/ P^m_X} \right| = \left| \frac{\Delta Q^S_X}{\Delta P_X} \right| \cdot \frac{P^m_X}{Q^{S.m}_X}$$

Percentage change of price

# Calculating Price Elasticity of Demand




If the price is measured in NTD

$$E_X^{D.P_X} = \left|\frac{200-100}{20-40}\right| \cdot \frac{30}{150} = 1$$

If the price is measured in D

$$E_X^{D.P_X} = \left|\frac{200-100}{0.5-1}\right| \cdot \frac{0.75}{150} = 1$$

# Demand Elasticity

- **If $Q \equiv Q_X^D$ （the quantity demanded for product X）,**
  **then**

$$Q_X^D = f(P_X \mid P_Y, I, \dots)\text{ ，}$$

**Price elasticity of demand：** $E_X^{D.P_X} = \left|\frac{\Delta Q_X^D / Q_{X.m}^D}{\Delta P_X / P_{X.m}}\right| = \left|\frac{\Delta Q_X^D}{\Delta P_X}\right| \cdot \frac{P_{X.m}}{Q_{X.m}^D}$ ,

**Cross elasticity of demand：** $E_X^{D.P_Y} = \frac{\Delta Q_X^D / Q_{X.m}^D}{\Delta P_Y / P_{Y.m}} = \frac{\Delta Q_X^D}{\Delta P_Y} \cdot \frac{P_{Y.m}}{Q_{X.m}^D}$ ,

**Income elasticity of demand：** $E_X^{D.I} = \frac{\Delta Q_X^D / Q_{X.m}^D}{\Delta I / I_m} = \frac{\Delta Q_X^D}{\Delta I} \cdot \frac{I_m}{Q_{X.m}^D}$ ,

where $Q_{X.m}^D = \frac{Q_{X.u}^D + Q_{X.d}^D}{2}, P_{X.m} = \frac{P_{X.u} + P_{X.d}}{2}, I_m = \frac{I_u + I_d}{2}$

- **If product X and Y are <u>substitute</u> and other things being equal,**

$$\boldsymbol{P_Y \downarrow, Q_Y^D \uparrow \Longrightarrow Q_X^D \downarrow.}$$

**Then the cross elasticity of demand**

$$E_X^{D.P_Y} = \frac{\Delta Q_X^D}{\Delta P_Y} \cdot \frac{P_{Y.m}}{Q_{X.m}^D} > 0.$$

- **If product X and Y are <u>complement</u> and other things being equal,**

$$\boldsymbol{P_Y \downarrow, Q_Y^D \uparrow \Longrightarrow Q_X^D \uparrow.}$$

**Then the cross elasticity of demand**

$$E_X^{D.P_Y} = \frac{\Delta Q_X^D}{\Delta P_Y} \cdot \frac{P_{Y.m}}{Q_{X.m}^D} < 0.$$

- **If product X is a normal good and other things being equal,**

$$\mathbf{I}\uparrow \Longrightarrow \boldsymbol{Q_X^D} \uparrow.$$

**Then the income elasticity of demand**

$$E_X^{D.I} = \frac{\Delta Q_X^D}{\Delta I} \cdot \frac{I_m}{Q_{X.m}^D} > 0.$$

- **If product X is a inferior good and other things being equal,**

$$\mathbf{I}\uparrow \Longrightarrow \boldsymbol{Q_X^D} \downarrow.$$

**Then the income elasticity of demand**

$$E_X^{D.I} = \frac{\Delta Q_X^D}{\Delta I} \cdot \frac{I_m}{Q_{X.m}^D} < 0.$$

# Supply Elasticity

- **If $Q \equiv Q_X^S$（the quantity supplied for product X）, then**

$$Q_X^S = g(P_X \mid P_Y, \dots)\ ,$$

**Price elasticity of supply：** $E_X^{S.P_X} = \left| \frac{\Delta Q_X^S / Q_{X.m}^S}{\Delta P_X / P_{X.m}} \right| = \left| \frac{\Delta Q_X^S}{\Delta P_X} \right| \cdot \frac{P_{X.m}}{Q_{X.m}^S},$

**Cross elasticity of supply：** $E_X^{S.P_Y} = \frac{\Delta Q_X^S / Q_{X.m}^S}{\Delta P_Y / P_{Y.m}} = \frac{\Delta Q_X^S}{\Delta P_Y} \cdot \frac{P_{Y.m}}{Q_{X.m}^S},$

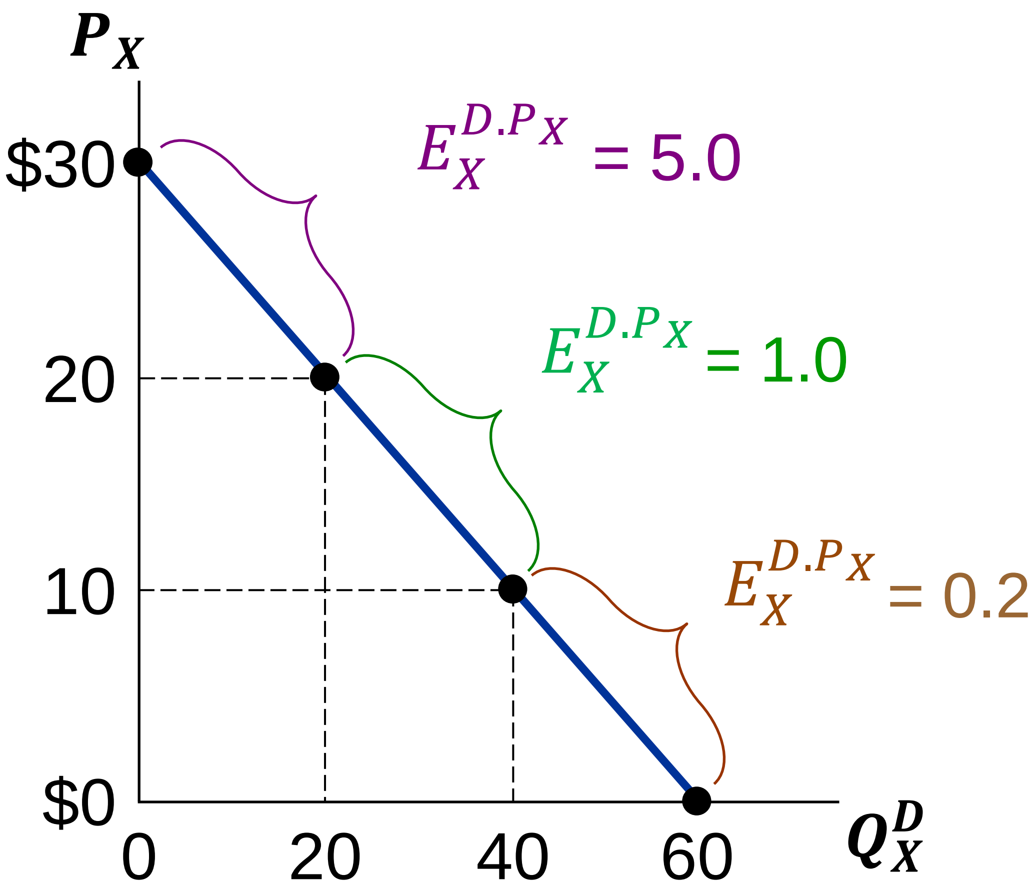
**where** $Q_{X.m}^S = \frac{Q_{X.u}^S + Q_{X.d}^S}{2},\ P_{X.m} = \frac{P_{X.u} + P_{X.d}}{2}$

Lecture3_W1-Q1

# 3-1.2: 供需曲線與價格彈性

**(Demand and Supply Curves and their Price Elasticities of Demand and Supply)**

# The price elasticity of demand on a linear demand curve




- **If the demand curve is linear, its slope is constant but the price elasticity of demand could be different**

- **When the price of the product is high, the price elasticity of demand is high**

**(Inference: As the price is high enough, buyers are more sensitive to the price change because they have more incentives to find substitutes.)**

# 需求有價格彈性、無價格彈性
(demand is elastic or inelastic)

- X 商品的需求有價格彈性(elastic)，當

$$E_X^{D.P_X} > 1$$ **(product X is elastic)**

- X 商品的需求沒有價格彈性(inelastic)，當

$$E_X^{D.P_X} < 1$$ **(product X is inelastic)**

- X 商品的需求具單位價格彈性(unit elastic)，當

$$E_X^{D.P_X} = 1$$ **(product X is unit elastic)**

# 供給有價格彈性、無價格彈性
(supply is elastic or inelastic)

- X 商品的供給有價格彈性(elastic)，當

$$E_X^{S.P_X} > 1$$ **(product X is elastic)**

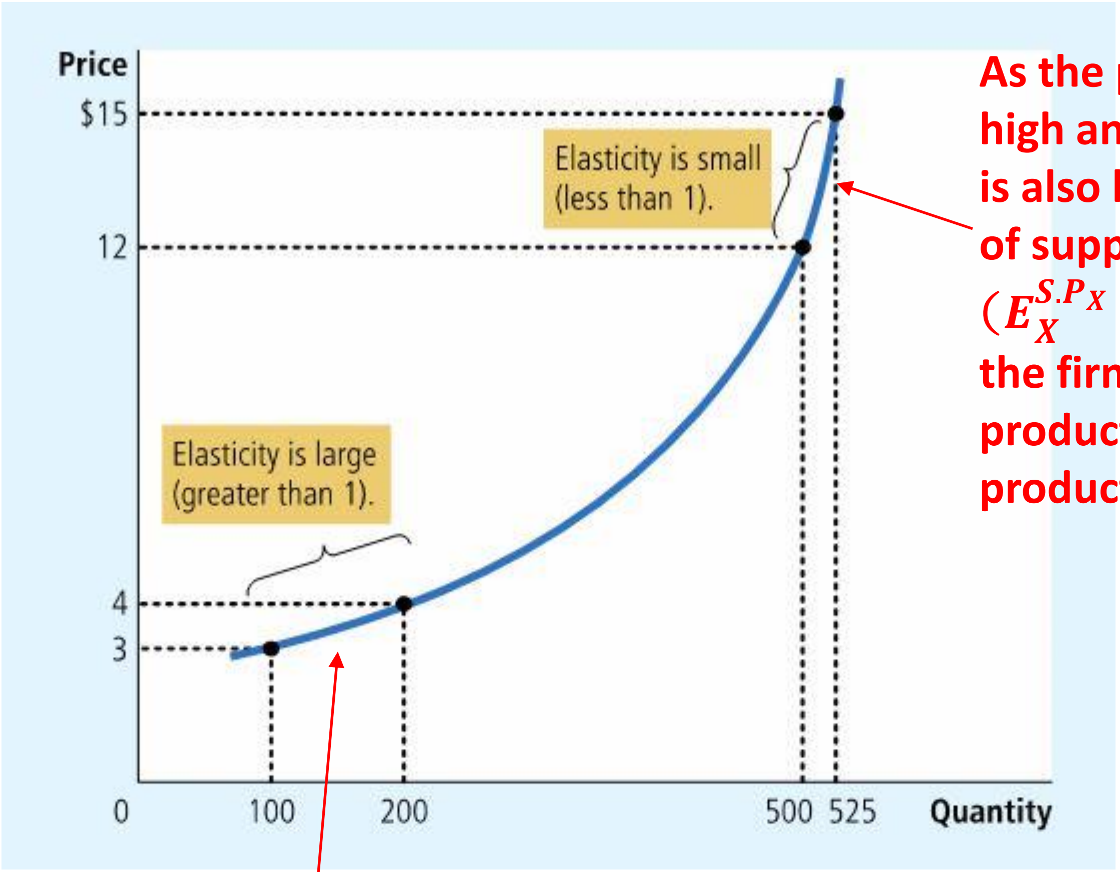
- X 商品的供給沒有價格彈性(inelastic)，當

$$E_X^{S.P_X} < 1$$ **(product X is elastic)**

- X 商品的供給具單位價格彈性(unit elastic)，當

$$E_X^{S.P_X} = 1$$ **(product X has unit elastic)**

– In reality, most supply curves are nonlinear




**As the price of the product is high and the product supplied is also high, the price elasticity of supply could be small ($E_X^{S.P_X} < 1$). That is because the firm does not have enough production capacity to raise its production.**

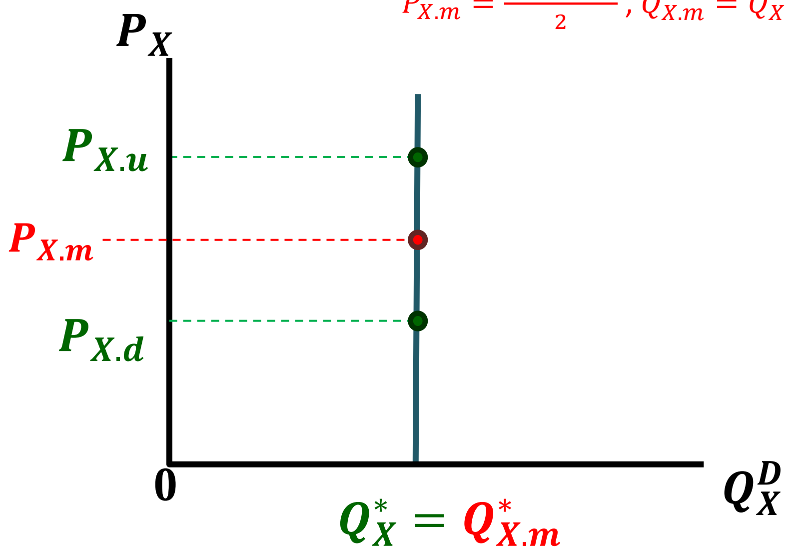
**As the price of the product is low and the product supplied is also low, the price elasticity of supply could be great ($E_X^{S.P_X} > 1$). That is because the firm have enough production capacity to raise its production.**

# 3-1.3:兩個供需曲線的特例

(Two Special Cases for Demand and Supply Curves and their Price Elasticities)

**Vertical D and S curves**

$P_{X.m} = \frac{P_{X.u}+P_{X.d}}{2}\,,\ Q^*_{X.m} = Q^*_X$

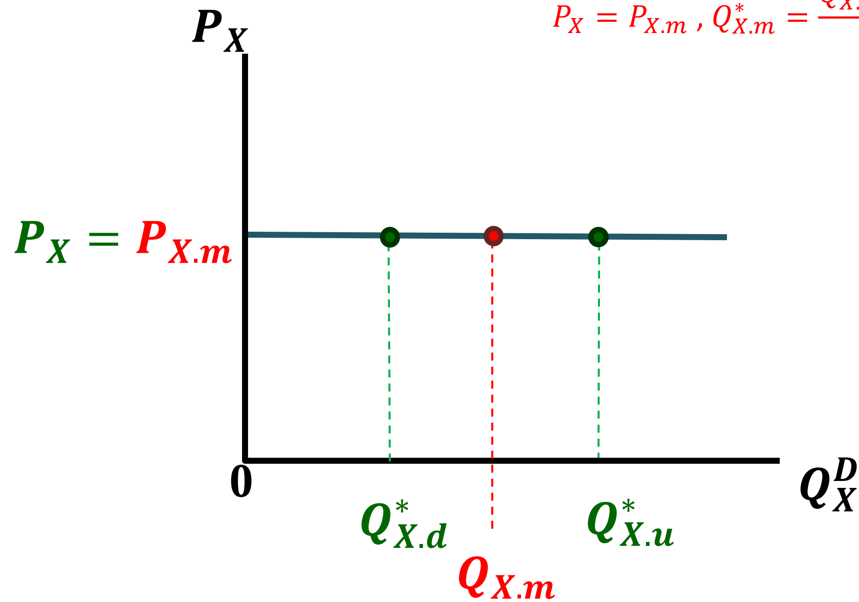



$$-E_X^{*.P_X} = \left|\frac{\Delta Q^*_X}{\Delta P_X}\right| \cdot \frac{P_{X.m}}{Q^*_{X.m}} = \left|\frac{1}{\Delta P_X/\Delta Q^*_X}\right| \cdot \frac{P^m_X}{Q^*_{X.m}} = \left|\frac{1}{\infty}\right| \cdot \frac{P_{X.m}}{Q^*_{X.m}} = 0$$

– As the D and S curves are vertical, the price elasticity is 0, says perfect inelasticity.

**Horizontal D and S curves**

$P_X = P_{X.m}\,,\ Q^*_{X.m} = \frac{Q^*_{X.u} + Q^*_{X.d}}{2}$




$$-E_X^{*.P_X} = \left|\frac{\Delta Q^*_X}{\Delta P_X}\right| \cdot \frac{P_{X.m}}{Q^*_{X.m}} = \left|\frac{1}{\Delta P_X/\Delta Q^*_X}\right| \cdot \frac{P_{X.m}}{Q^*_{X.m}} = \left|\frac{1}{0}\right| \cdot \frac{P_{X.m}}{Q^*_{X.m}} = \infty$$

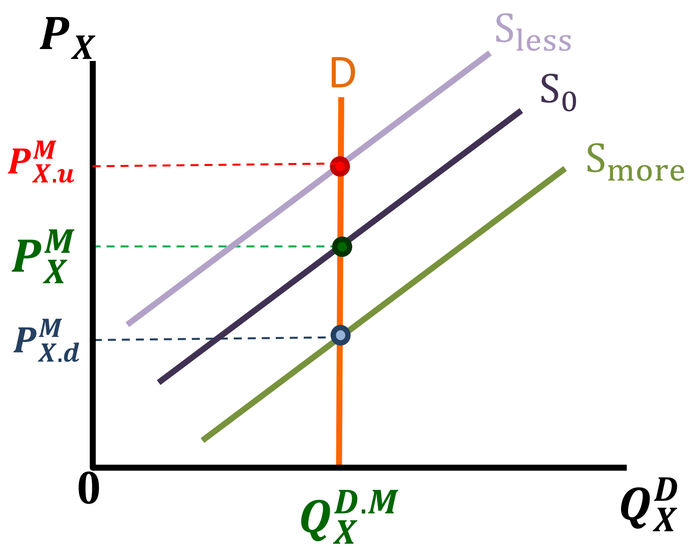
– As the D and S curves are horizontal, the price elasticity is infinite, says perfect elasticity.

- **Perfect (or Close to Perfect) Inelasticity**
  - The quantity demanded or supplied is (almost) indifferent to the change of price. In other words, the product could be any price given quantity demanded or supplied.

  - In real world, there are no product with perfect inelasticity but there are similar ones close to perfect inelasticity within given price interval

    E.g. special daily necessities, like salt and sugar；
    high addicted products, like drug or cigarette
    some special medical expenses, like medical expenses for Keratitis surgery

**If the demand curve is vertical**




(1) Suppose the market demand for product X is $Q_X^{DM}$ and the product is perfect inelasticity

(2) The price of the product will be totally determined by its market quantity supplied, like $P_X^M$

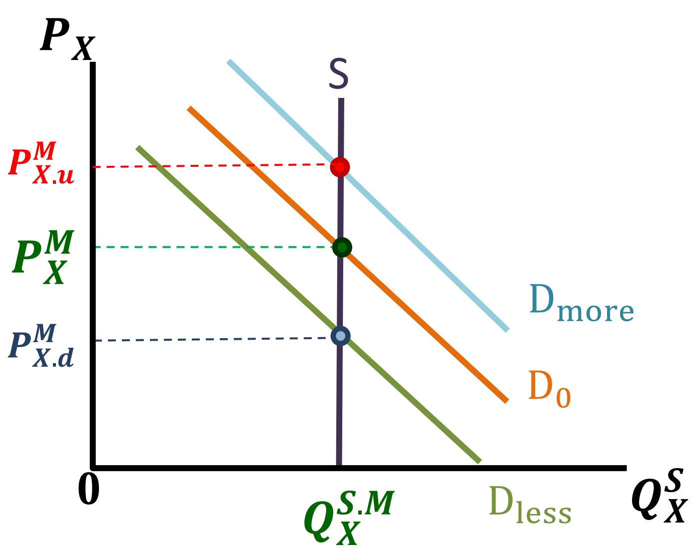
(3) If the quantity supplied decreases, the price will raise up to $P_{X.u}^M$

(4) If the quantity supplied increases, the price will decrease down to $P_{X.d}^M$

(5) It is a scenario of price totally determined by quantity supplied

(6) For example, drug is expensive because the ban on its providing.

**If the supply curve is vertical**




(1) Suppose the market supply for product X is $Q_X^{SM}$ and the product is perfect inelasticity

(2) The price of the product will be totally determined by its market quantity demanded, like $P_X^M$

(3) If the quantity demanded increases, the price will raise up to $P_{X.u}^M$

(4) If the quantity demanded decreases, the price will decrease down to $P_{X.d}^M$

(5) It is a scenario of price totally determined by quantity demanded

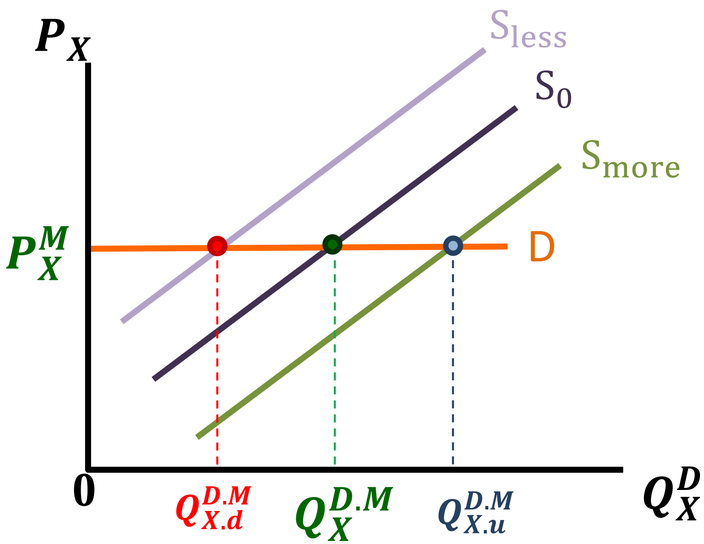
(6) For example，the fish catch per day

- **Perfect (or Close to Perfect) elasticity**
  - The price is (almost) indifferent to change of quantity demanded and supplied. In other words, the product could be any quantity demanded or supplied given a price.

  - In real world, there are no product with perfect elasticity. That is, no buyer but government can absorb any quantity provided by suppliers at a price, and no supplier can produce any quantity due to limited production capacity

E.g. government can lock the exchange rate between one currency to another currency.

**If the demand curve is horizontal**



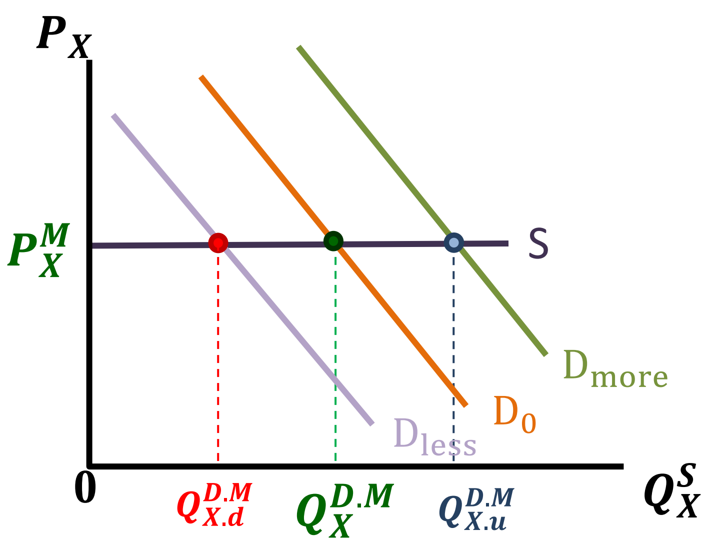

(1) Suppose the market demand for product X is $Q_X^{DM}$ and the product is perfect elasticity

(2) The product will always be bought at $P_X^M$

(3) It is a scenario of quantity totally determined by suppliers

(4) The buyer could be the government, and the supplier could be farmers; the product X could be crops or exchange rate

**If the supply curve is horizontal**




(1) Suppose the market price for product X is $P_X^M$ and the product is perfect elasticity

(2) No matter how high the price is, the buyers need accept this price the suppliers propose

(3) It is a scenario of quantity totally determined by buyers

(4) The supplier could be a multinational corporation, and the buyers could be the people in a small country. Their quantity demanded accounts for merely a small portion of the quantity the corporation supplied.

# 需求價格彈性小結論

(A brief conclusion on price elasticity of demand)

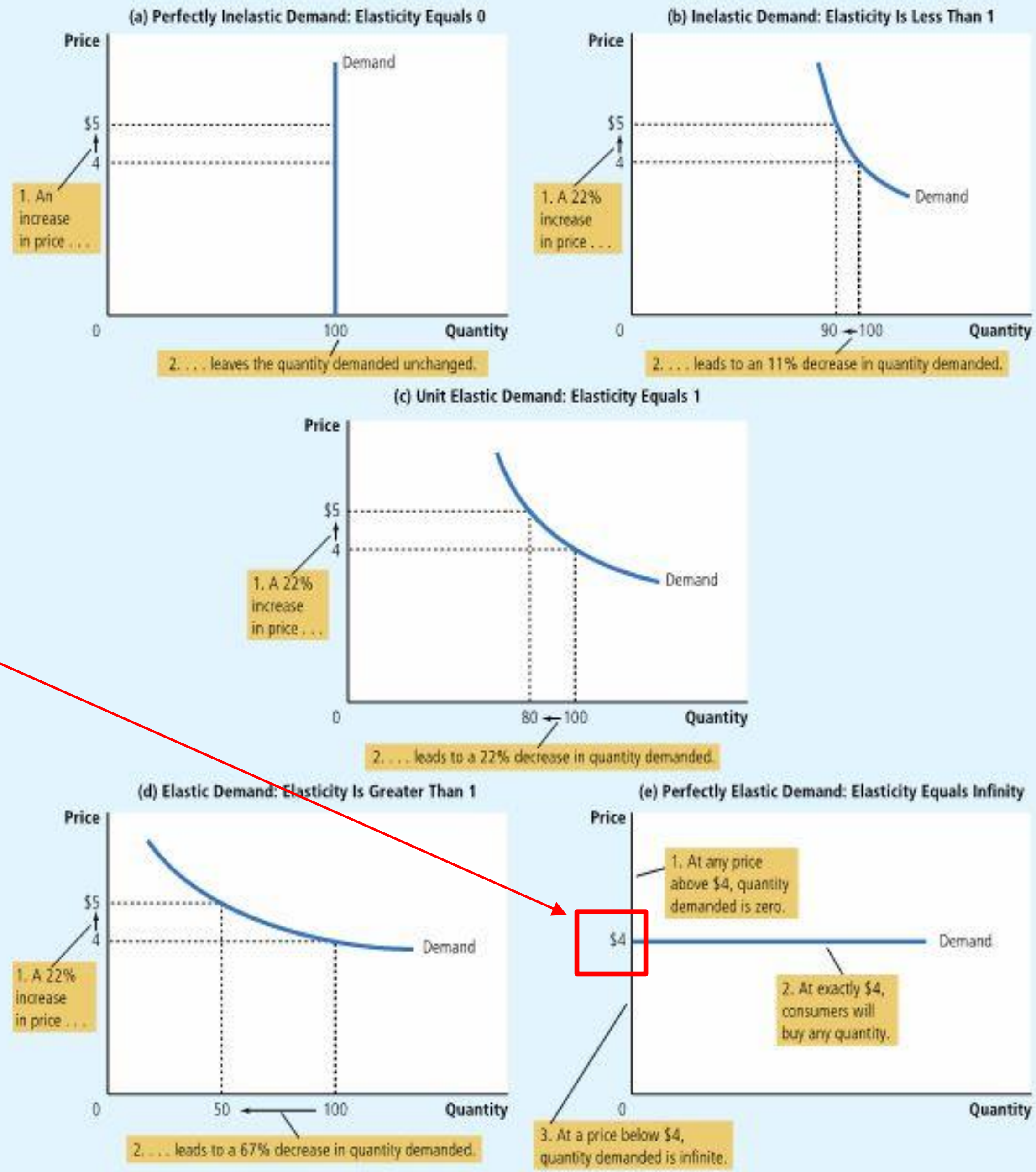
**$4 是消費者願意購買最高價格，所以如果需求曲線為水平，則在該價格時，消費者願意接受任意產量；該價格以上則消費者不願購買；該價格以下，消費者的願意購買量為無限大**

**(If the price is below $4, the quantity demanded is infinite. Otherwise, the quantity demanded is 0)**

**FIGURE 1**

**The Price Elasticity of Demand**

The price elasticity of demand determines whether the demand curve is steep or flat. Note that all percentage changes are calculated using the midpoint method.



# A Few Price Elasticities of Demand from the Real World

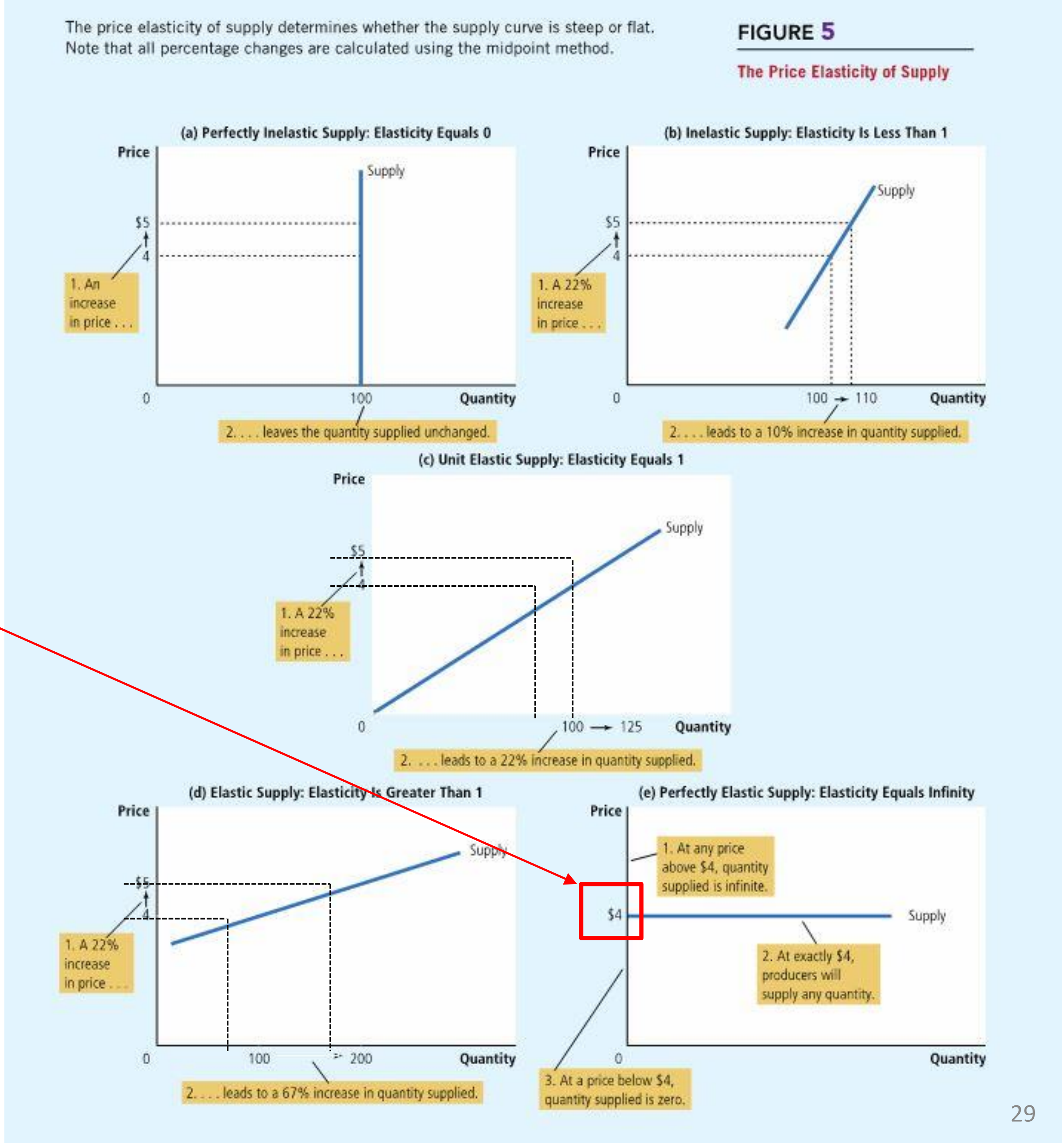
| | |
|---|---|
| Eggs | 0.1 |
| Healthcare | 0.2 |
| Cigarettes | 0.4 |
| Rice | 0.5 |
| Housing | 0.7 |
| Beef | 1.6 |
| Peanut Butter | 1.7 |
| Restaurant Meals | 2.3 |
| Mountain Dew | 4.4 |

# 供給價格彈性
# 小結論

(A brief conclusion on price elasticity of supply)

**$4 是廠商願意生產最低價格，所以如果供給曲線為水平，則在該價格時，廠商願意生產任意產量；該價格以下廠商的生產量是0；該價格以上，廠商的生產量是無限大**

**(If the price is below $4, the quantity supplied is 0. Otherwise, the quantity demanded is infinite)**

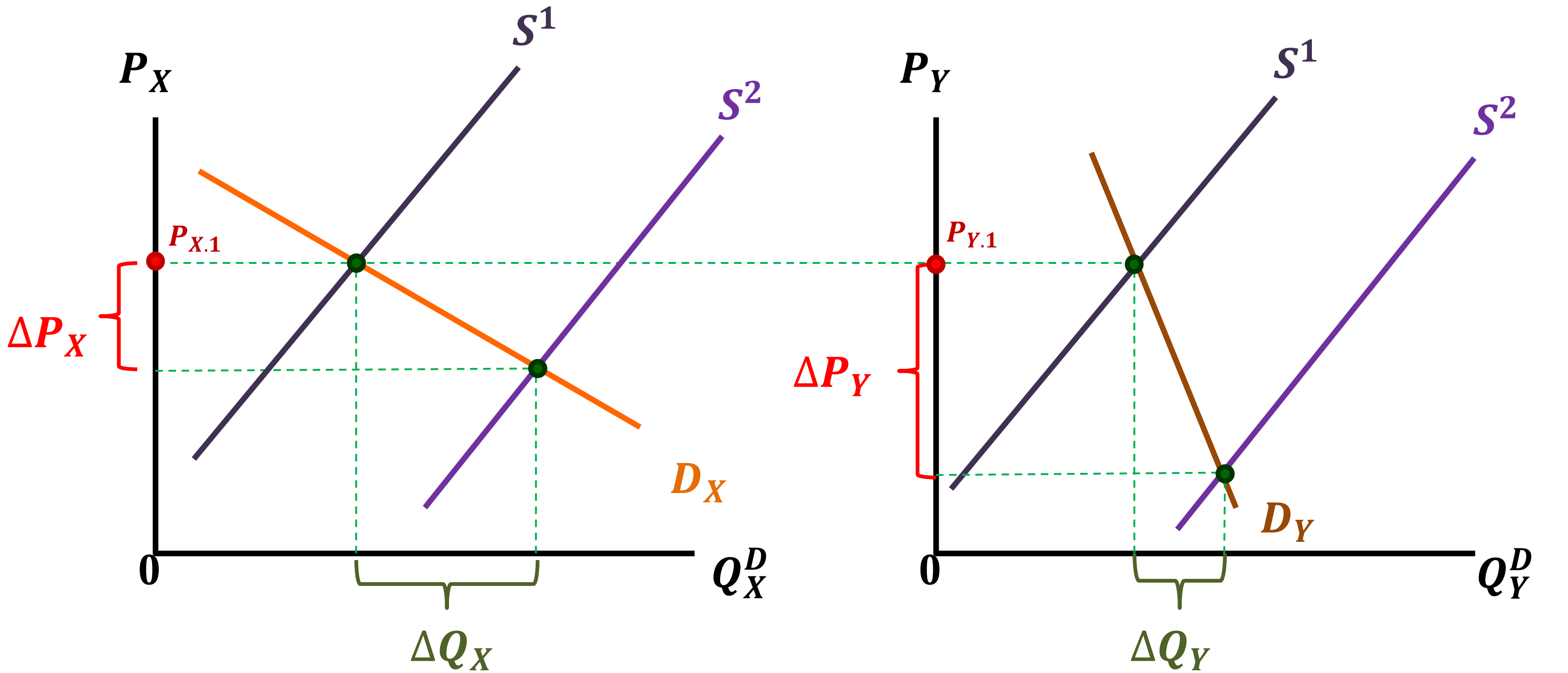


# 3-2.1: 彈性大小與均衡變動

(the relation between elasticity and equilibrium)

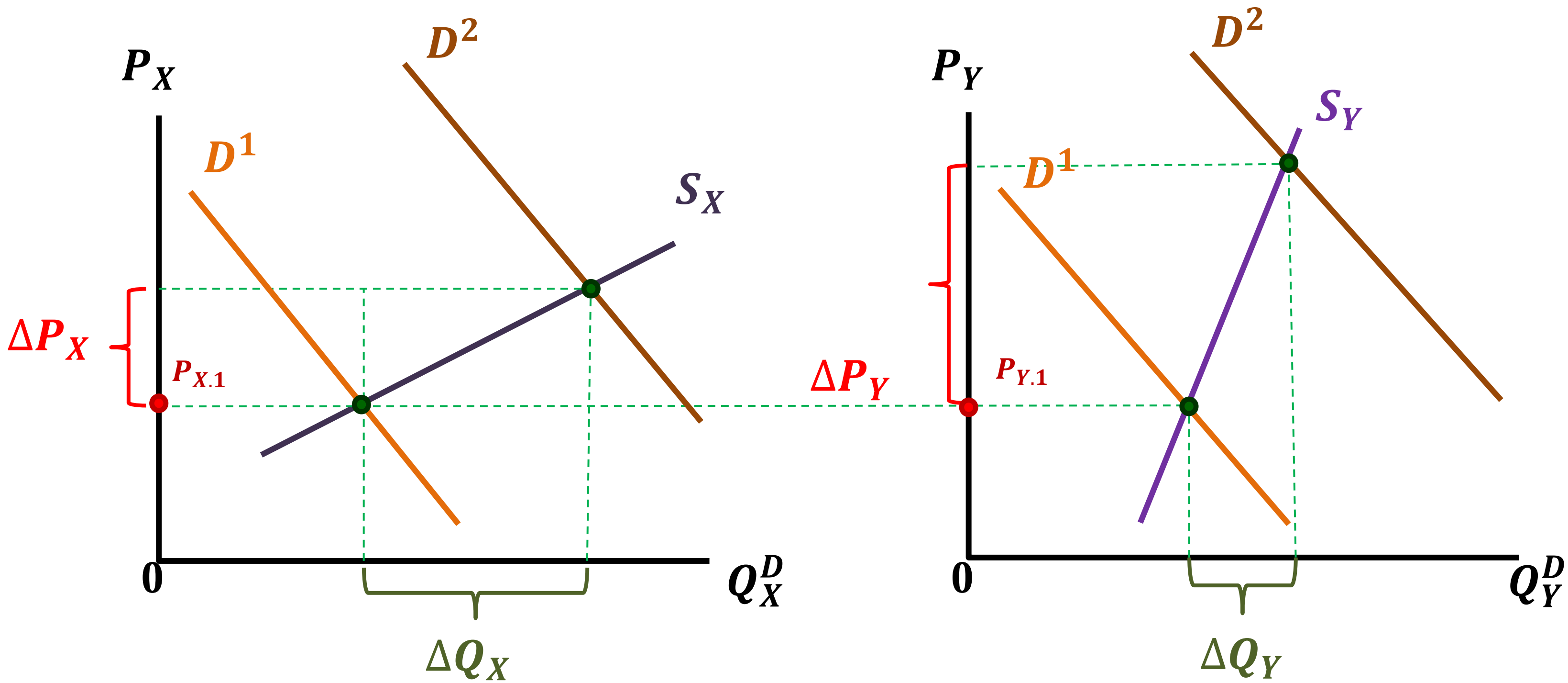
- 以下我們做<u>比較靜態分析</u>
  **(In this section, we perform comparative static analyses)**
  - 當需求或供給曲線因為種種「其他條件」改變而左右移動時，市場均衡價、量也會跟著改變
    **(As the demand or supply curve shift because one of the shifters changes, the equilibrium quantity and price will also change)**
  - 探討市場均衡如何因其他條件改變而變動的研究，稱為比較靜態分析
    **(We here perform comparative statics to investigate how the equilibrium is affected by the changes of demand and supply shifters)**

# Price elasticity of demand ($E^{D.P}$) and equilibrium




- **Two demand curves：$D_X$ 、 $D_Y$, where $\left|{}^{\Delta Q_X}/_{\Delta P_X}\right| > \left|{}^{\Delta Q_Y}/_{\Delta P_Y}\right|$ and $E_X^{D.P_X} > E_Y^{D.P_Y}$**
- **As the supply curve moves from $S^1$ to $S^2$, $\Delta P_X < \Delta P_Y$ and $\Delta Q_X > \Delta Q_Y$**
- **Given the initial price $P^* = P_{X.1} = P_{Y.1}$ and if the suppliers decide to increase certain volume of quantity supplied, the price of the product with high $E^{D.P}$ decreases less than that with low $E^{D.P}$; the quantity demanded of the product with high $E^{D.P}$ increases more than that with low $E^{D.P}$**

# Price elasticity of supply ($E^{S.P}$) and equilibrium




- **Two supply curves: $S_X$ 、 $S_Y$, where $\left|{}^{\Delta Q_X}/_{\Delta P_X}\right| > \left|{}^{\Delta Q_Y}/_{\Delta P_Y}\right|$ and $E_X^{S.P_X} > E_Y^{S.P_Y}$**
- **As the demand curve moves from $D^1$ to $D^2$ ，$\Delta P_X < \Delta P_Y$ and $\Delta Q_X > \Delta Q_Y$**
- **Given the initial price $P^* = P_{X.1} = P_{Y.1}$ and if the buyers decide to increase certain volume of quantity demanded, the price of the product with high $E^{S.P}$ increases less than that with low $E^{S.P}$; the quantity demanded of the product with high $E^{S.P}$ increases more than that with low $E^{S.P}$**

# 小結論
## (Brief Conclusion)

- 大致而言，當需求或供給價格彈性大，則均衡點的改變大多反應在均衡數量，價格反應較小
  (Once the product has high price elasticity of demand or supply, the equilibrium quantity demanded or supplied is more sensitive than equilibrium price as the supply or demand curve shift)

- 反之，當需求或供給價格彈性小，則均衡點的改變大多反應在均衡價格，數量反應較小
  (Otherwise, the equilibrium quantity demanded or supplied is more sensitive than equilibrium price as the supply or demand curve shift)

# 3-2.2: 決定價格彈性大小的因素

(The Determinants of Price Elasticities)

- **The determinants of Price Elasticity of Demand:**

**(a) The numbers of substitutes**

the greater the number of substitute, the greater the price elasticity of demand;

e.g., the cell phones with Android system

milk and soy milk

**(b) The percentage of your total consumption the product accounts for**

the smaller the percentage, the smaller the price elasticity of demand;

e.g., salt or soy-bean sauce

**(c) The preference and loyalty to the product**

the greater the loyalty, the smaller the price elasticity of demand;

e.g., the fans of Iphone

**(d) The length of time span**

the longer the length of time span, the greater the price elasticity of demand because the buyers have more time to find substitutes;

e.g., rental suites

- **The determinants of Price Elasticity of Supply:**

  **(a)Whether or not the production factor could be used to produce another products**

  If the production factor is widely used for producing other products, the price elasticity of supply could be great

  **(b) The increment of production costs as the quantity supplied increases**

  If the production cost increases sharply as the quantity supplied increases, the suppliers are less willing to greatly increase their quantity supplied. Thus, the greater the increment in production cost as the quantity supplied increases, the smaller the price elasticity of supply

  **(c) The length of time span**

  the longer the length of time span, the greater the price elasticity of supply because the suppliers have more time to improve the efficiency of their production lines

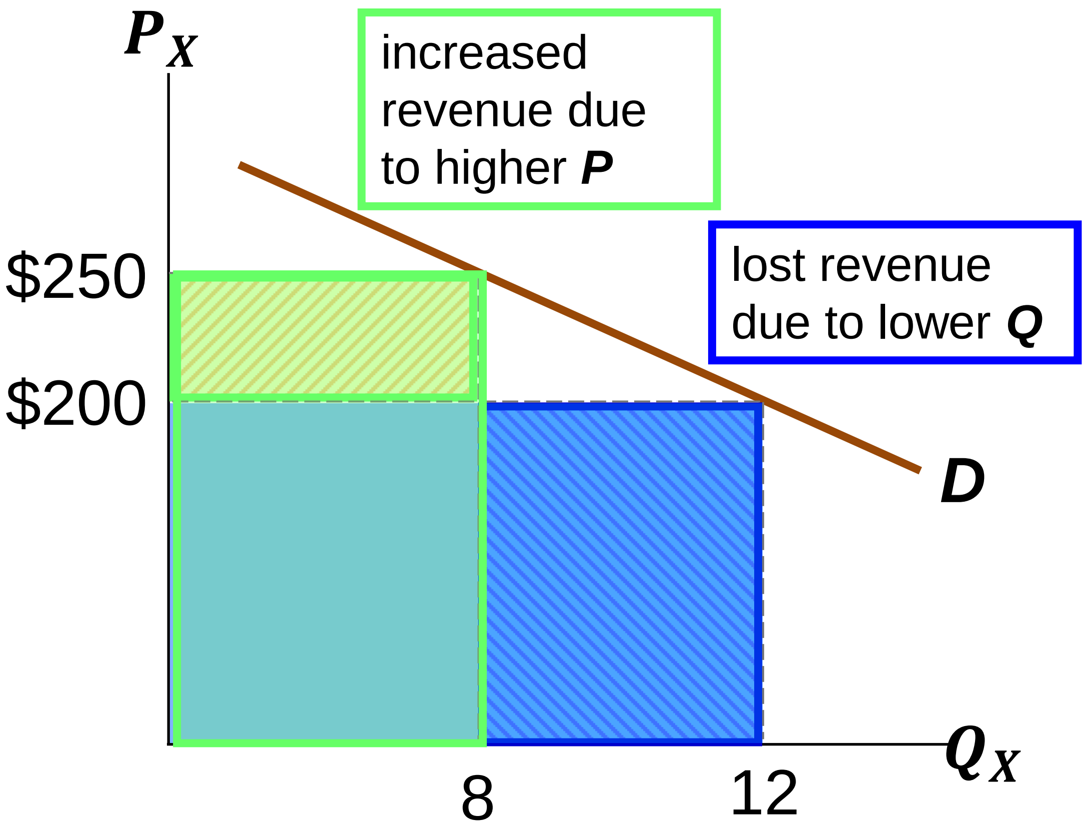
# 3-2.3:
# 從消費者的需求曲線來看
# 需求價格彈性的大小與生產者總收益的關係

(relation between price elasticity of demand and suppliers' total revenue)

先定義X商品生產者的總收益:

(Total Revenue, TR) = $P_X \times Q_X$

**(Definition of Suppliers' Total Revenue of Product X)**

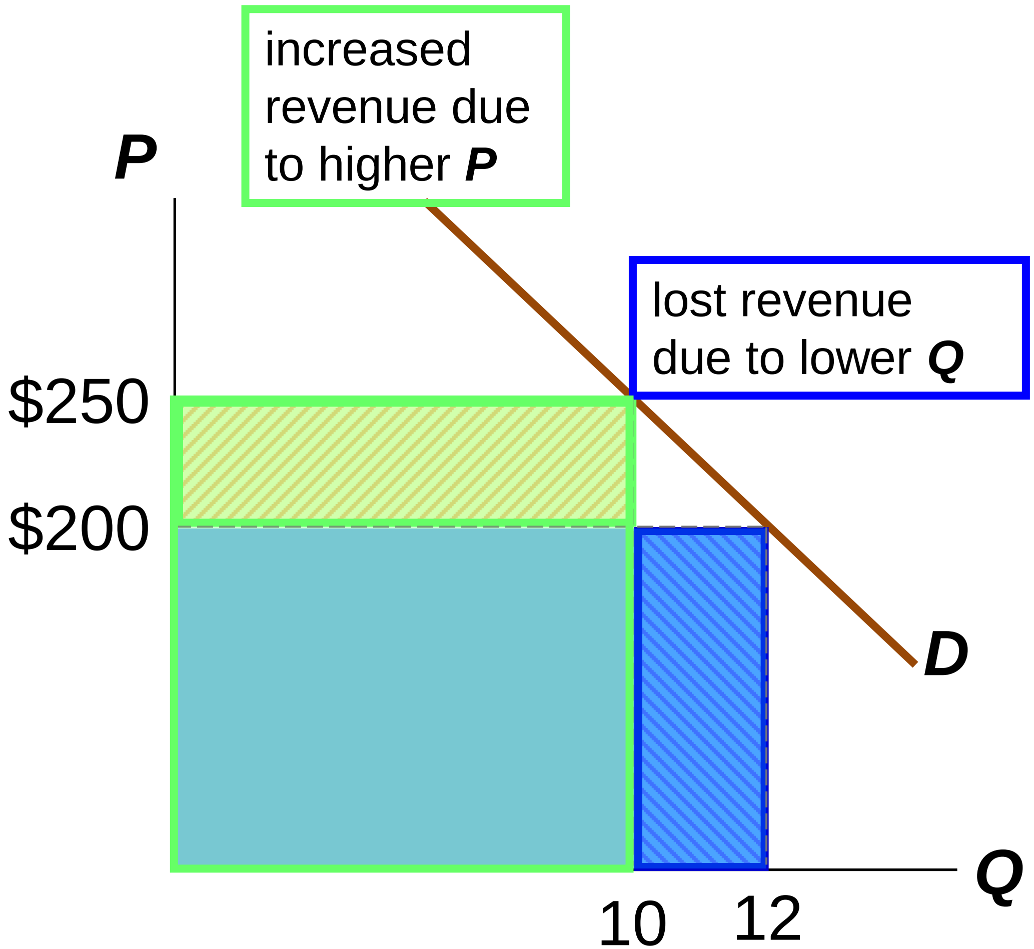



- $E_X^{D.P_X}$ = 1.8

- If ***P*** = $200, ***Q*** = 12, and TR = $2400

- If ***P*** = $250, ***Q*** = 8, and TR = $2000

**When the product *X* is elastic, a increase in price causes TR to fall.**




- $E_X^{D.P_X}$ = 0.82

- If ***P*** = $200, ***Q*** = 12, and TR =$2400

- If ***P*** = $250, ***Q*** = 10, and TR = $2500

**When the product *X* is inelastic, a price increase causes TR to rise.**

# 3-2.4: 兩個實例分析
(Two Examples)

# 實例 1.1 – 以價制量

(Case 1.1: Could price suppresses quantity demanded?)

- 早年油品、菸酒由政府壟斷，統一定價

(In Taiwan, gasolines and cigarette are provided merely by a government-sponsored monopolistic firm in early years. Their prices are determined by the firm.)

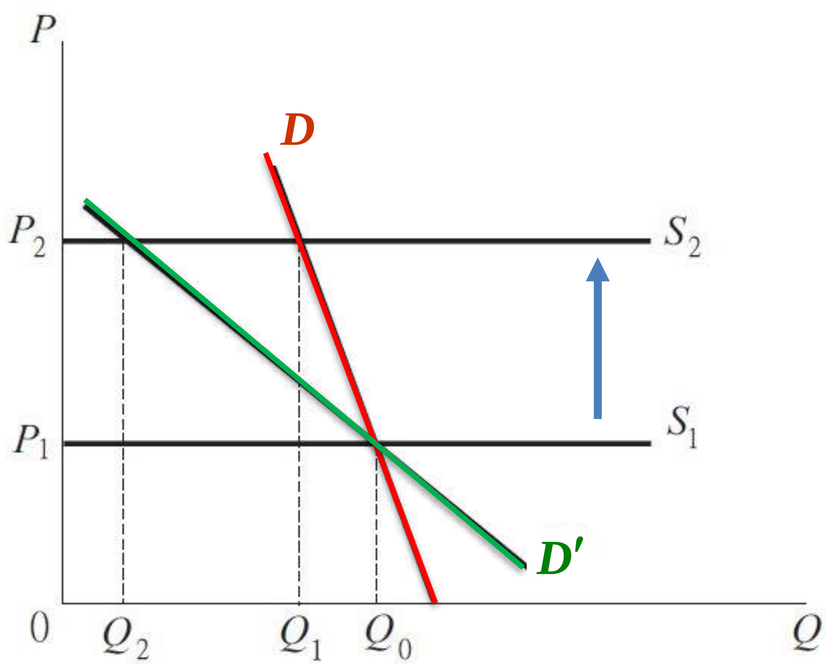
- 政府的理由：

耗用汽油浪費能源，菸酒不利人民健康，由政府統一訂定高價，以抑制消費

(The government states that high price can save the energy resources. Besides, high prices could prevent people from buying cigarette doing hurt to health)

- 在此我們暫不討論以上的道德邏輯是否合理，只探討「高定價是否真能抑制消費」

(Could high prices suppress quantity demanded?)




- If the monopolistic firm is the only supplier, it can determine the price of the product. Thus, the supply curve is horizontal.

  (the price elasticity of supply is infinite – the buyers have no choices but accept the price)

If the initial price is $P_1$, the firm decides to raise the price up to $P_2$

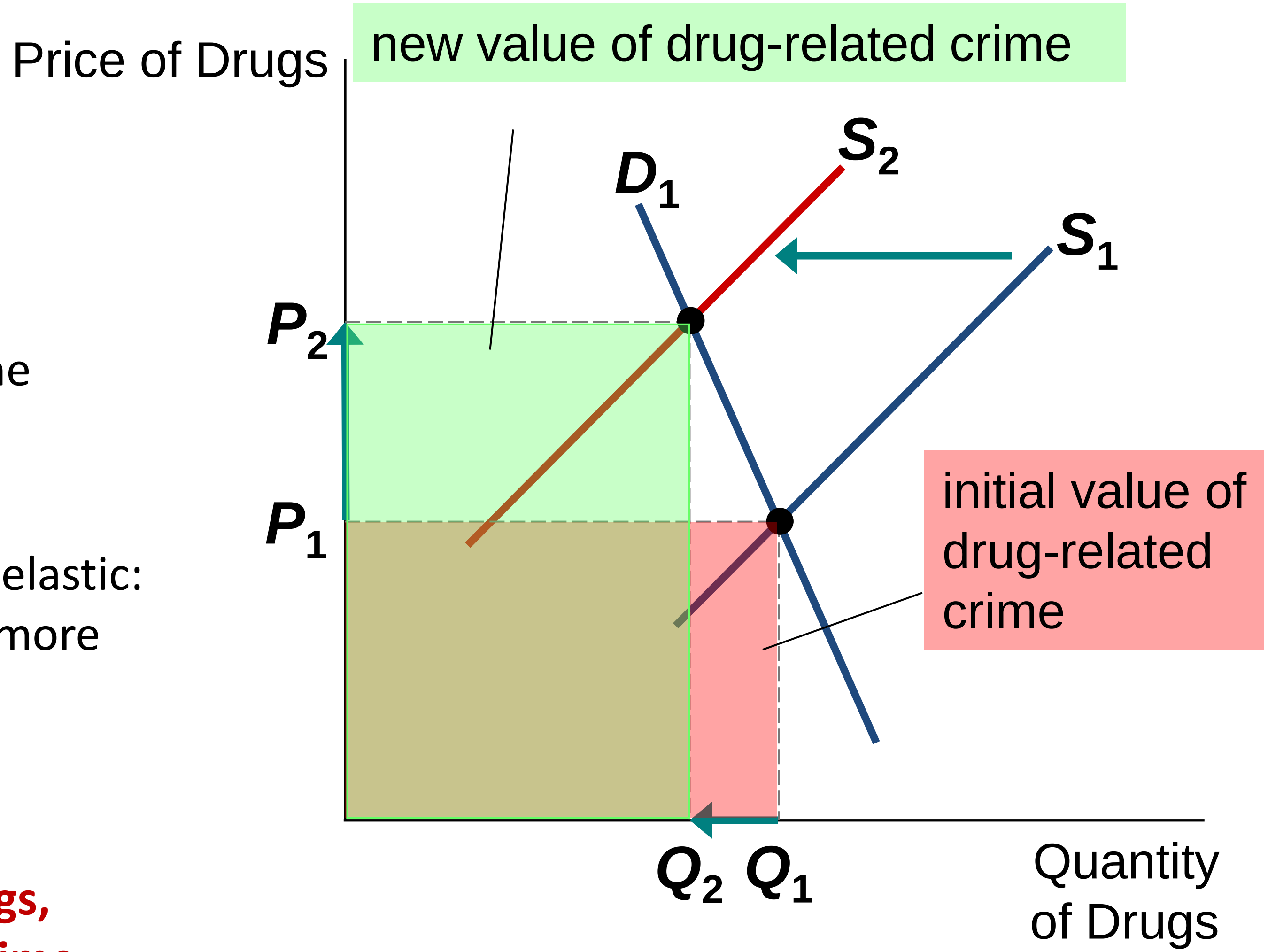
If the demand curve implies inelasticity, increasing price cannot suppress quantity demanded well; if the demand curve implies elasticity, raising price can suppress quantity demanded well

# Case 1.2: Does Drug Interdiction Increase or Decrease Drug-related Crime?

1. **Increase the number of federal agents devoted to the war on drugs**
   - Illegal drugs: supply curve shifts left
     - Higher price and lower quantity
   - Amount of drug-related crimes
     - Inelastic demand for drugs
     - Higher drugs price: higher total revenue
     - Increase drug-related crime

# Policy 1: Interdiction

- Interdiction reduces the supply of drugs.
- Demand for drugs is inelastic: ***P*** rises proportionally more than ***Q*** falls.
- **Result: an increase in total spending on drugs, and in drug-related crime**



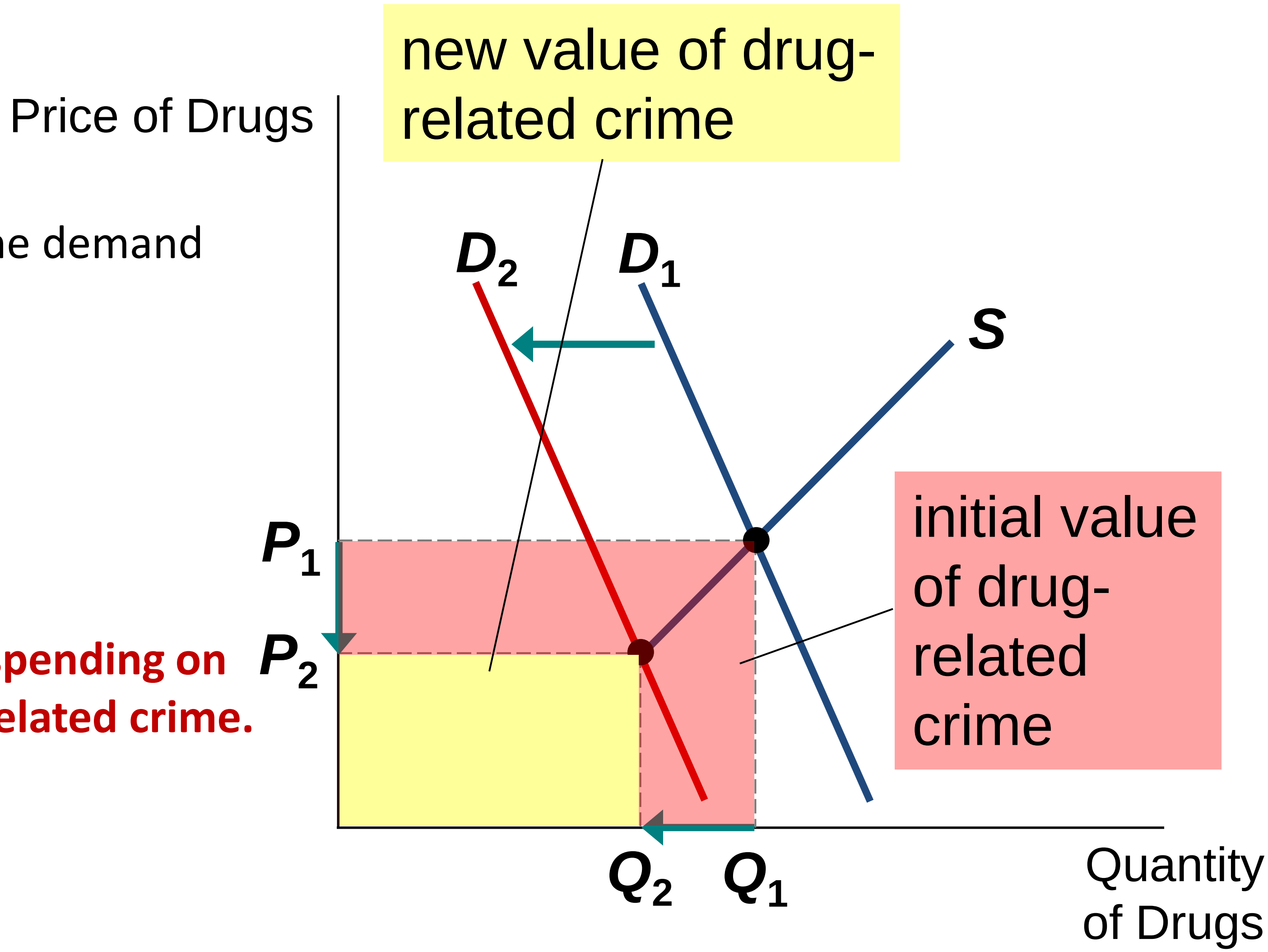
# Does Drug Interdiction Increase or Decrease Drug-related Crime?

## 2. Policy of drug education

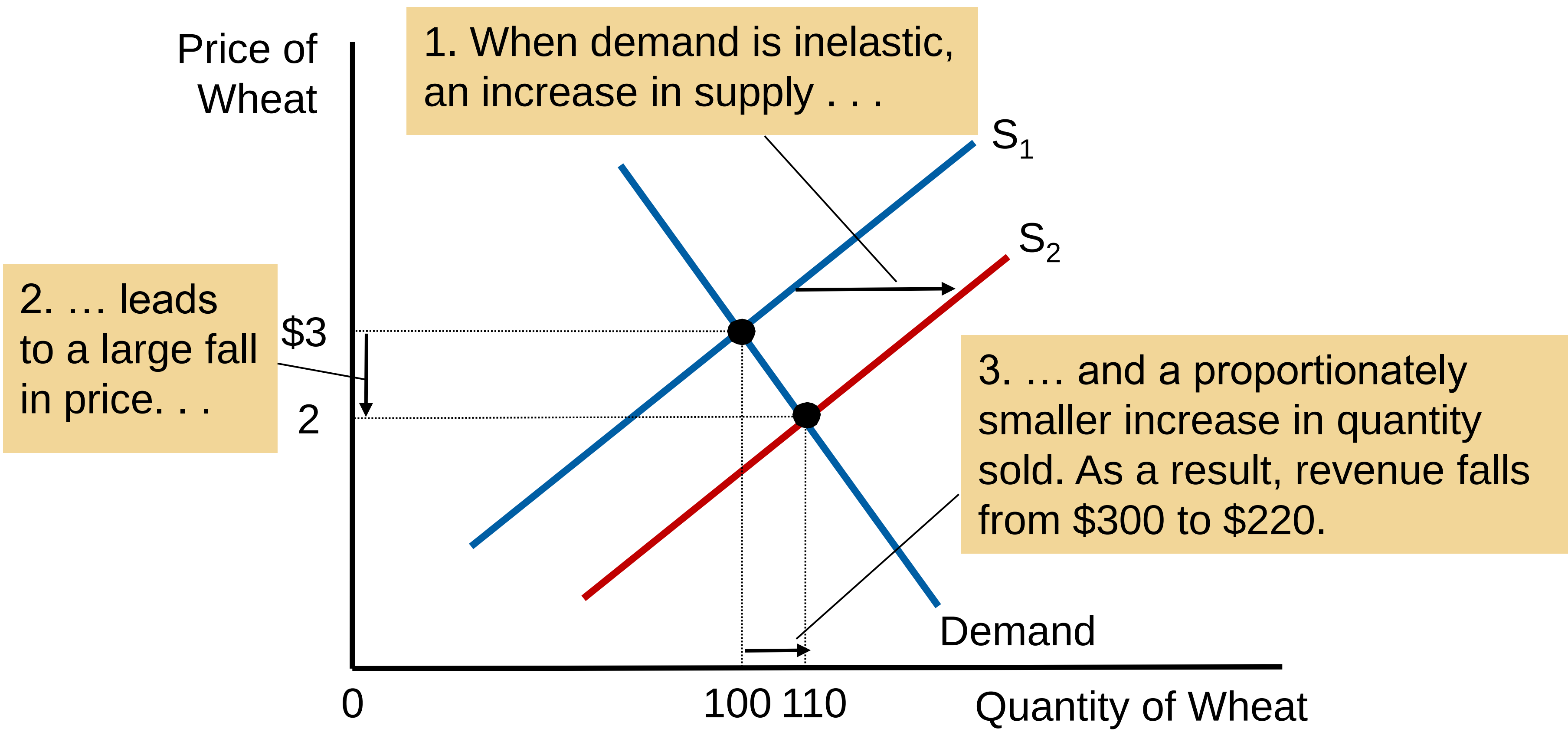
- Reduce demand for illegal drugs
- Left shift of demand curve
- Lower quantity
- Lower price
- Reduce drug-related crime

# Policy 2: Education

- Education reduces the demand for drugs.
- ***P*** and ***Q*** fall.
- **Result:**
  **A decrease in total spending on drugs, and in drug-related crime.**

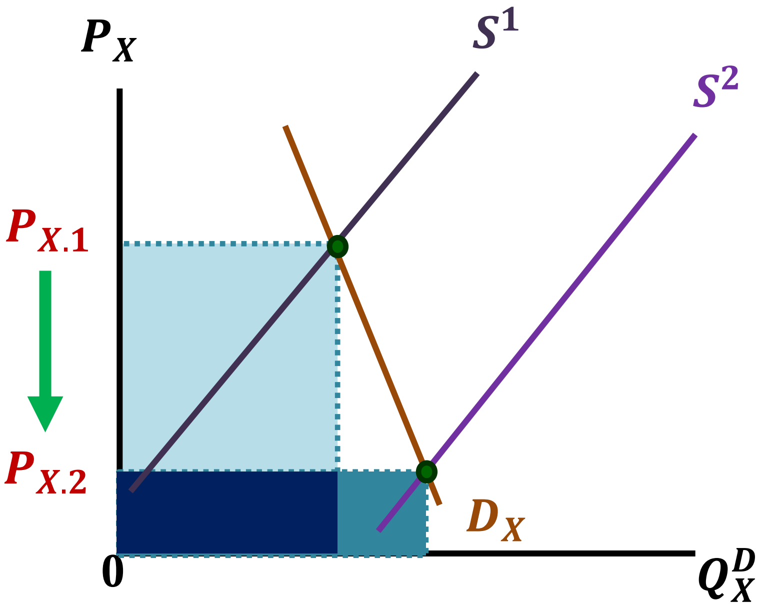


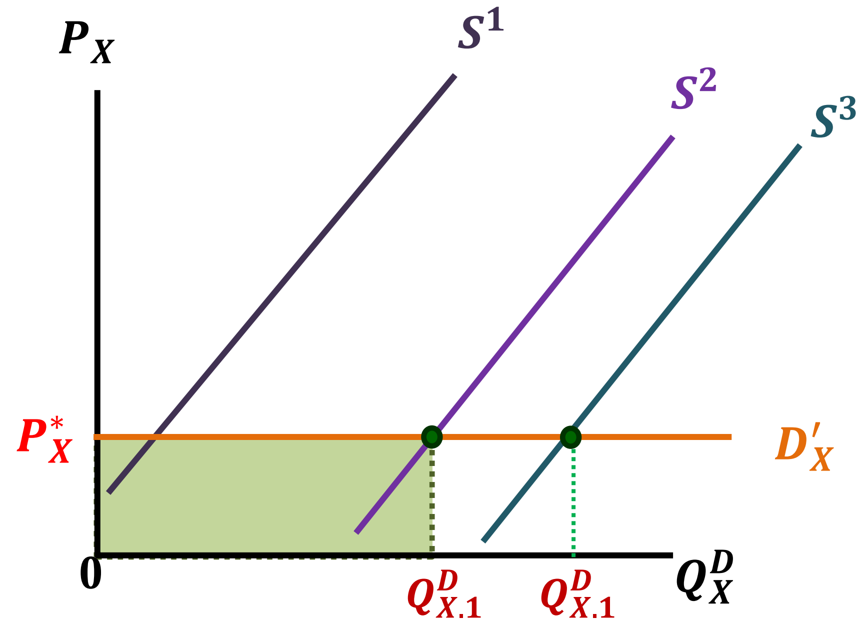
# An Increase in Supply in the Market for Wheat



# Case 1.3: Government absorbs all of the crops at a guaranteed price

E_Lecture3_W2-Q2




- Good news for farming shifts the supply curve to the right, and that makes the price of the crop decrease from $P_{X.1}$to $P_{X.2}$. For the crop are inelastic product. Decrement in price results in decrement in TR.
- If the government buy all of the residual crops at a guaranteed price $P_X^*$, $P_X^* > P_{X.2}$, then the demand curve of the crop becomes horizontal

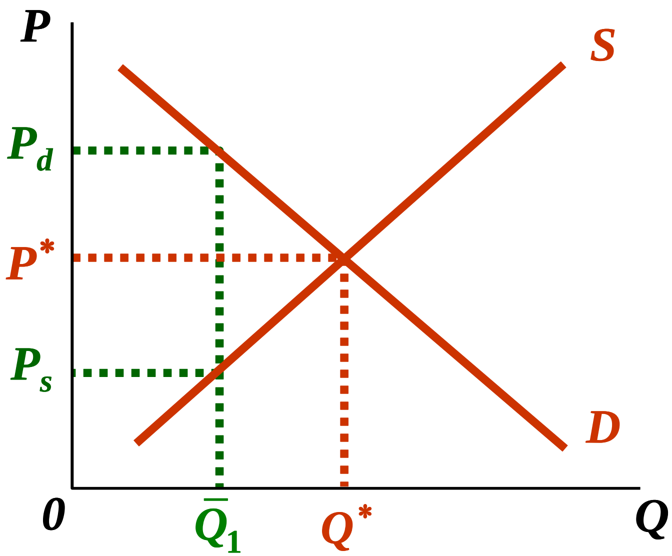
# Case 2.1 – Restriction on quantity demanded and supplied

- Government can impose restriction on quantity demanded or supplied to the volume $\bar{Q}$, , like restriction on water usage amount.

- If the equilibrium quantity is below $\bar{Q}$, then the restriction is useless.

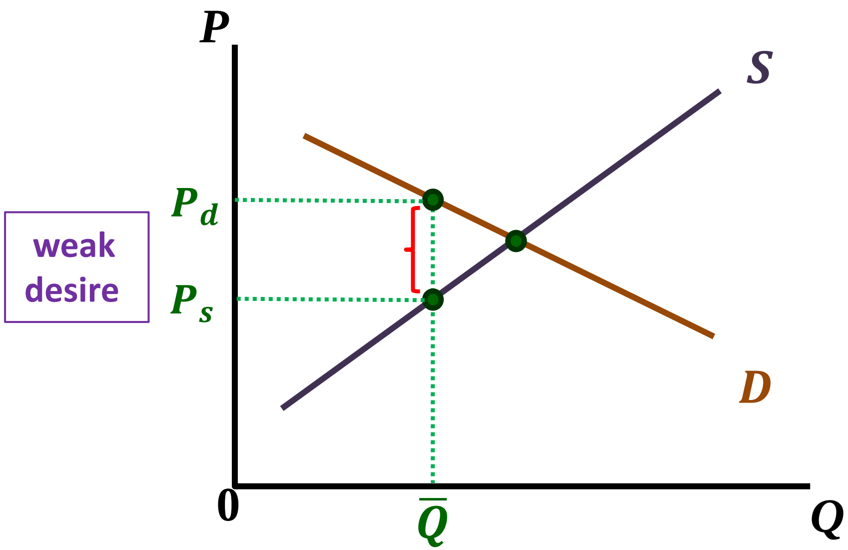
- If the equilibrium quantity is above $\bar{Q}$, then this restriction may influence the equilibrium.

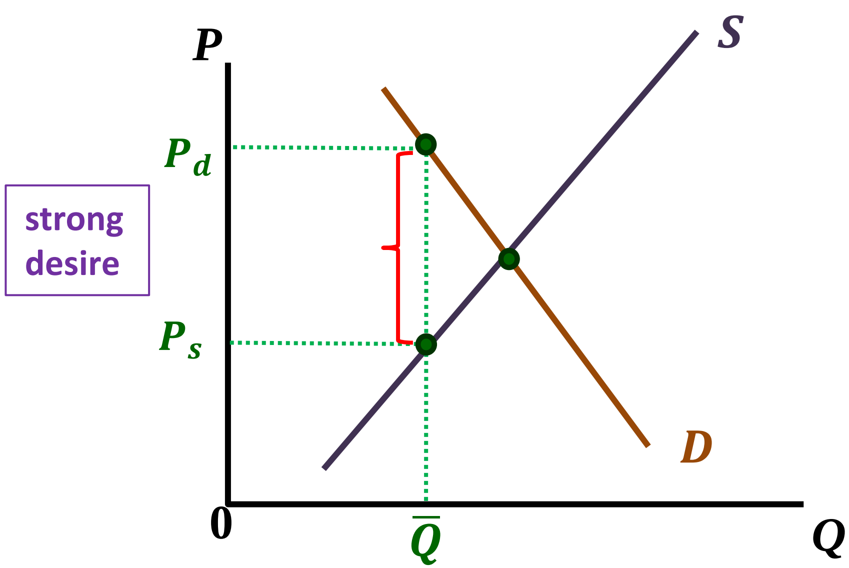
# Restriction on Quantity and Black Market




- Before imposing restriction, the equilibrium quantity and price of the product are $P^*$ and $Q^*$
- Now the quantity restriction is imposed to be $\bar{Q}_1$
- If the quantity transacted is up to $\bar{Q}_1$, the market need to be closed. The corresponding price $P_{s,}$ the lowest price the sellers accept, is lower than $P_d$, the highest price the buyer accept. That means both of the buyers and sellers still desire to make transactions
- However, the presence of the restriction encourage the occurrence black markets.

# The desire for transaction with the presence of restriction



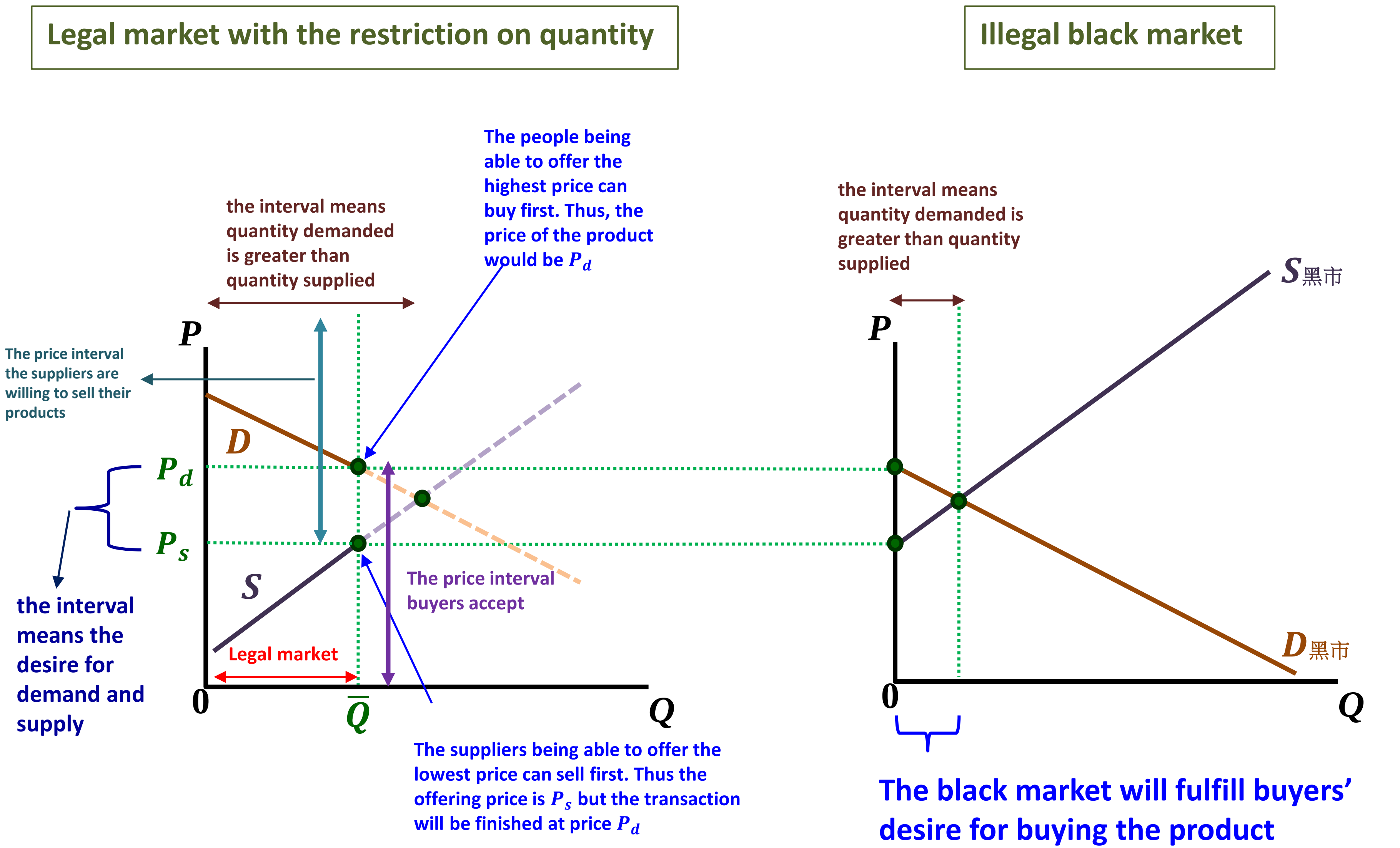




- Restriction on quantity does not always suppress the desire for buying or selling the products.
- The less the price elasticity, the greater the difference in ($P_d$ - $P_s$ ).
- That means the presence of restriction on the quantity leads to stronger desire for transaction.

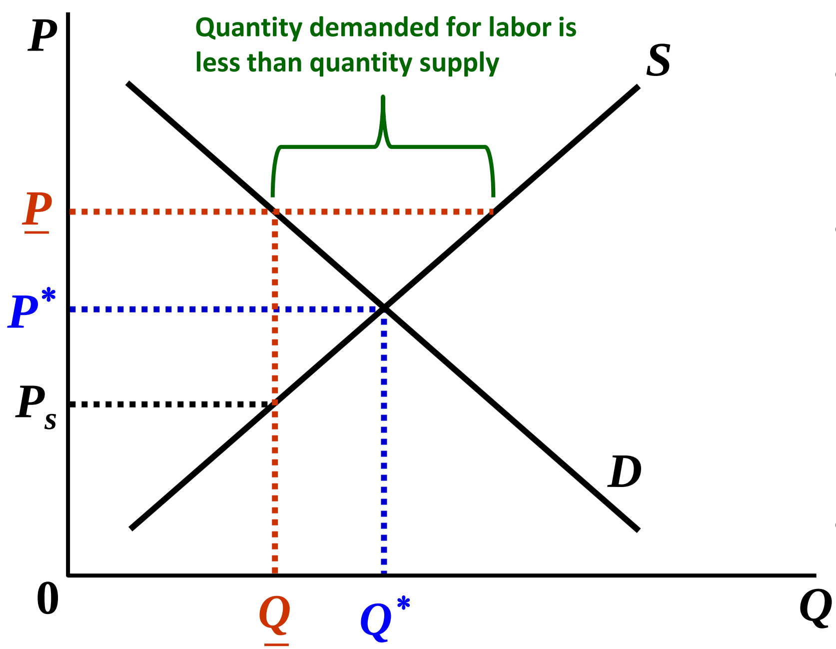
# The demand and supply curve in the black market

- 假設政府採取限量管制，由出價高的先買，售價低的先賣。如果買不到的人會轉入黑市，則黑市中的供給與需求曲線如下:

(Suppose that the government imposes restriction on quantity, and in the market the people being able to offer the highest price can buy the product first and the suppliers being able to offer the lowest price can sell their product first. Once the products are unavailable in the legal market, people desiring to get it will go to the black market. The labeled diagrams of the demand and supply in the black market will be…)


Legal market with the restriction on quantity
Illegal black market
The people being able to offer the highest price can buy first. Thus, the price of the product would be $P_d$
the interval means quantity demanded is greater than quantity supplied
the interval means quantity demanded is greater than quantity supplied
$S_{黑市}$
P
P
The price interval the suppliers are willing to sell their products
D
$P_d$
$P_s$
S
The price interval buyers accept
the interval means the desire for demand and supply
Legal market
0
$\overline{Q}$
Q
0
Q
$D_{黑市}$
The suppliers being able to offer the lowest price can sell first. Thus the offering price is $P_s$ but the transaction will be finished at price $P_d$
The black market will fulfill buyers' desire for buying the product

# Case 2.2 – Restriction on Price




- The government sets the minimum guaranteed salary $\underline{P}$ for per unit of labor
- If $\underline{P}$ is less than the equilibrium salary $P*$, the restriction is useless.
- If the $\underline{P}$ is greater than $P*$, then the quantity demanded for labor will less than quantity supplied, leading to unemployment.
- To avoid unemployment once $\underline{P}$ is set and $\underline{P} > P*$, the government authority concerned need provide ($\underline{P}$ - $P*$) to employers as allowance.